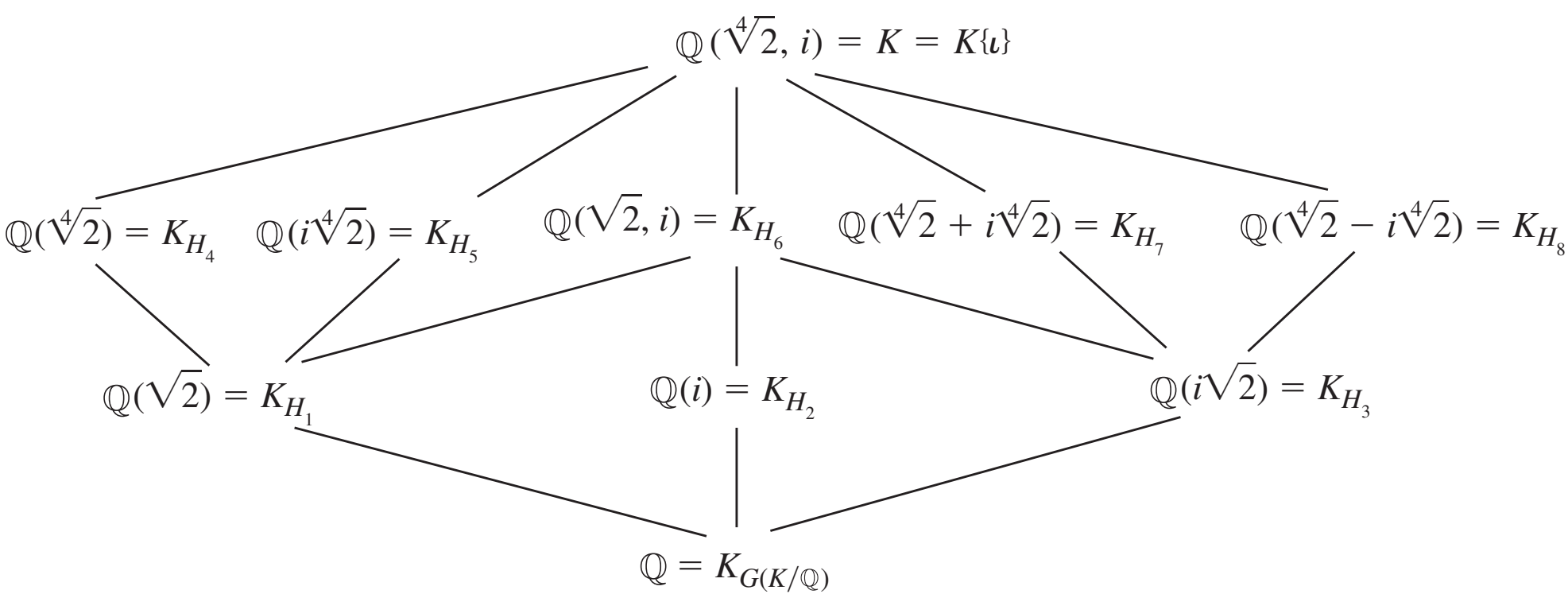

$\mathbb{Q}(\sqrt[4]{2}, i) = K = K\{\iota\}$
$\mathbb{Q}(\sqrt[4]{2}) = K_{H_4}$
$\mathbb{Q}(i\sqrt[4]{2}) = K_{H_5}$
$\mathbb{Q}(\sqrt{2}, i) = K_{H_6}$
$\mathbb{Q}(\sqrt[4]{2} + i\sqrt[4]{2}) = K_{H_7}$
$\mathbb{Q}(\sqrt[4]{2} - i\sqrt[4]{2}) = K_{H_8}$
$\mathbb{Q}(\sqrt{2}) = K_{H_1}$
$\mathbb{Q}(i) = K_{H_2}$
$\mathbb{Q}(i\sqrt{2}) = K_{H_3}$
$\mathbb{Q} = K_{G(K/\mathbb{Q})}$

(b)